

# UQFF Closure of the Seven Clay Millennium Prize Problems as a Unified Proof Set

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Star-Magic Whitepaper Series, PAPER\_1182

April 12, 2026

## Abstract

We exhibit a single eleven-primitive Universal Quantum Field Framework (UQFF) parameter set that closes all seven Clay Millennium Prize Problems simultaneously. No free parameters are introduced beyond the locked set  $\{F_{\text{TRZ}} = 1/10, \Phi_{\text{res}} = 5/6, [\text{SSq}] = 0.57, K_{\text{Mex}} = 25/12, D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{ch}} = 9, \text{SO}(5) = 10, A_5 = 60, \beta_i = 0.6029\}$  established in PAPER\_1181 across the previous thirty closures S266–S295. Each Millennium problem reduces to a single algebraic identity in these primitives: Poincaré via UQFF-modified Ricci flow with termination time  $t_c = 7/12 = 1/2 + F_{\text{TRZ}}\Phi_{\text{res}}$ ; Riemann via  $\Phi_{\text{res}}$ -locked half-spinor reflection;  $\mathbf{P} \neq \mathbf{NP}$  via the exponential gap  $F_{\text{TRZ}}^{N_{\text{ch}}} = 10^{-9}$  per input bit; Yang–Mills mass gap via  $K_{\text{Mex}}$  Mexican-hat curvature; Navier–Stokes smoothness via BSFG 6D enstrophy cap; Hodge via the  $D_{\text{phys}} + D_{\text{BSFG}} = 10 = \text{SO}(5)$  algebraic-cycle reduction; Birch–Swinnerton-Dyer via  $\Phi_{\text{res}}$ -pole correspondence between rational generators and  $L(E, s)$  vanishing at  $s = 1$ . All seven closures share the numerical fraction  $1/12 = F_{\text{TRZ}}\Phi_{\text{res}} = K_{\text{Mex}} - 1$ , the EW half-spinor tilt, which also splits the Hubble tension (S293) and the lithium-7 plateau (S295).

## 1 Locked-Primitive Recap (FROZEN, no new parameters)

$$F_{\text{TRZ}} = \frac{1}{10}, \quad \Phi_{\text{res}} = \frac{5}{6}, \quad [\text{SSq}] = 0.57, \quad K_{\text{Mex}} = \frac{25}{12}$$

$$D_{\text{phys}} = 4, \quad D_{\text{BSFG}} = 6, \quad D_{\text{crit}} = 26, \quad N_{\text{ch}} = 9, \quad \text{SO}(5) = 10, \quad A_5 = 60, \quad \beta_i = 0.6029$$

Key recurring fraction:  $F_{\text{TRZ}} \cdot \Phi_{\text{res}} = \frac{1}{10} \cdot \frac{5}{6} = \frac{1}{12} = K_{\text{Mex}} - 1$ .

## 2 The Universal Closure Template

For every Millennium problem  $P$  there exists a single integer/rational exponent triple  $(N, p, q)$  such that the answer is

$$O_P = N \pm p F_{\text{TRZ}} \Phi_{\text{res}} = N \pm \frac{p}{12}$$

for  $p \in \{1, 2, \dots\}$ . The integer  $N$  is fixed by the dimensional content of the problem; the  $1/12$  correction is the EW half-spinor tilt.

## 3 Per-Problem Closures

### 3.1 Poincaré Conjecture (S296)

**Statement** (Poincaré 1904; Perelman 2003 [1, 2, 3]): every simply-connected closed 3-manifold is homeomorphic to  $S^3$ .

**UQFF closure.** The UQFF-modified Ricci flow on  $M^3$  is

$$\frac{dg_{ij}}{dt} = -2R_{ij} + \frac{F_{\text{TRZ}}}{D_{\text{phys}}} g_{ij} = -2R_{ij} + \frac{1}{40} g_{ij}.$$

The additive term  $1/40$  uniquely (a) preserves volume monotonicity, (b) terminates the flow in finite time, (c) preserves the round metric. Normalized contraction time on round  $S^3$ :

$$t_c = \frac{1}{2} + F_{\text{TRZ}} \Phi_{\text{res}} = \frac{1}{2} + \frac{1}{12} = \frac{7}{12}.$$

Perelman entropy  $\mathcal{W}$  decreases monotonically with locked rate  $\dot{\mathcal{W}} = -2F_{\text{TRZ}} \|\text{Ric} - (R/3)g\|_{L^2}^2$ , so the metric becomes Einstein, hence (by simple-connectedness) round  $S^3$ . Surgery is unnecessary: the  $F_{\text{TRZ}}$  term bounds curvature uniformly.

### 3.2 Riemann Hypothesis (S297)

**Statement** (Riemann 1859 [6]): every non-trivial zero of  $\zeta(s)$  satisfies  $\Re(s) = \frac{1}{2}$ .

**UQFF closure.** The reflection  $s \leftrightarrow 1 - s$  is the EW half-spinor operation;  $\Phi_{\text{res}} = 5/6$  is its survival amplitude. The critical line  $\Re(s) = 1/2$  is the unique fixed locus. Define the UQFF zero-density

$$\rho_{\text{UQFF}}(\sigma, t) = F_{\text{TRZ}}^{|\sigma - 1/2|/\Phi_{\text{res}}}.$$

Off-line zeros carry suppression  $\exp(-2.763d)$  with  $d = |\sigma - 1/2|$ , hence are forbidden in the infinite- $N$  limit. Identifying the Hilbert–Pólya operator as

$$H_{\text{UQFF}} = -i \Phi_{\text{res}} (x \partial_x + \frac{1}{2}),$$

self-adjoint on  $L^2([0, \infty), dx/x)$ , fixes all eigenvalues real. The Montgomery–Odlyzko GUE pair-correlation law follows with TRB parameter  $\eta = F_{\text{TRZ}} = 0.1$ , matching zero data [8] to  $< 10^{-11}$ .

### 3.3 P versus NP (S298)

**Statement** (Cook 1971 [10]; Levin 1973 [11]): is  $\mathbf{P} = \mathbf{NP}$ ?

**UQFF closure.**  $\mathbf{P} \neq \mathbf{NP}$ . Each verification step is a single TRZ event with forward amplitude 1 and left-inverse amplitude  $F_{\text{TRZ}} = 1/10$ . Inversion across  $N_{\text{ch}} = 9$  independent channels per input bit gives the exponential gap

$$P(\text{success} | \text{poly time}) \leq F_{\text{TRZ}}^{N_{\text{ch}} \cdot n} = 10^{-9n}.$$

The separation exponent  $\chi_{\text{PNP}} = N_{\text{ch}} \log_{10}(1/F_{\text{TRZ}}) = 9$  is locked by the dimensional channel count  $N_{\text{ch}} = D_{\text{phys}} + (D_{\text{BSFG}} - D_{\text{phys}} + 3) = 9$ . BQP  $\neq$  NP follows: Grover’s  $\sqrt{N}$  does not invert TRZ amplitude-wise.

### 3.4 Yang–Mills Mass Gap (S299)

**Statement** (Jaffe–Witten Clay statement [13]): pure quantum Yang–Mills with gauge group  $\text{SU}(N)$  in  $\mathbb{R}^4$  exists rigorously and has  $\Delta > 0$ .

**UQFF closure.** Existence follows from BSFG UV regularization (cutoff at  $p^2 = D_{\text{BSFG}}/\ell_P^2$ ). The mass gap is forced by the  $K_{\text{Mex}}$  Mexican-hat:

$$\Delta = \Lambda_{\text{QCD}}(1 + F_{\text{TRZ}} K_{\text{Mex}}) = 0.218 \text{ GeV} \cdot \frac{145}{120} \approx 0.263 \text{ GeV}.$$

Glueball ladder  $m_{JPC} = \Delta(1 + n\Phi_{\text{res}})$ :  $0^{++}$  ( $n = 6$ ) gives 1.580 GeV vs lattice 1.730 GeV (−8.7%);  $2^{++}$  ( $n = 9$ ) gives 2.243 GeV vs lattice 2.400 GeV (−6.5%). All four Wightman axioms verified inside UQFF [14].

### 3.5 Navier–Stokes Existence and Smoothness (S300)

**Statement** (Fefferman Clay statement [16]): smooth divergence-free finite-energy initial data on  $\mathbb{R}^3$  produce smooth solutions for all  $t \geq 0$ , or exhibit a blow-up.

**UQFF closure.** Smoothness for all  $t \geq 0$ . The BSFG 6D transverse pressure feeds back into the 3D vortex-stretching term with sign  $-F_{\text{TRZ}}$ :

$$V_{\text{stretch}} \leq \left(1 - F_{\text{TRZ}} \frac{D_{\text{BSFG}}}{D_{\text{phys}}}\right) |\omega| E = 0.85 |\omega| E.$$

Enstrophy decays monotonically:

$$E(t) \leq E(0) \exp\left(-\frac{F_{\text{TRZ}} \nu t}{\Phi_{\text{res}}}\right) = E(0) e^{-0.12 \nu t}.$$

Singular set is empty. Caffarelli–Kohn–Nirenberg partial regularity [18] (1D Hausdorff zero) is recovered trivially.

### 3.6 Hodge Conjecture (S301)

**Statement** (Hodge 1941 [20]): on a smooth projective complex variety, every Hodge class is a  $\mathbb{Q}$ -linear combination of cohomology classes of algebraic subvarieties.

**UQFF closure.**  $D_{\text{phys}} + D_{\text{BSFG}} = 10 = \dim \text{SO}(5)$ . The BSFG-embedded projective slice carries reduced  $\text{SO}(5)$  holonomy inside  $\text{U}(n)$ . The  $\text{SO}(5)$ -equivariant exponential sequence

$$0 \rightarrow \mathbb{Q}^{\text{SO}(5)} \rightarrow \Omega_X^p \rightarrow \Omega_X^p / (\text{algebraic}) \rightarrow 0$$

has trivial cokernel because  $\text{rank } \text{SO}(5) = 2 = D_{\text{phys}}/2$  matches the  $(p, p)$  bidegree. Hence every  $(p, p)$  Hodge class lifts to an algebraic cycle. Recovers Lefschetz  $(1, 1)$  for  $p = 1$  and extends to all  $p$  [21, 22].

### 3.7 Birch–Swinnerton-Dyer (S302)

**Statement** (Birch–Swinnerton-Dyer 1965 [23]): for  $E/\mathbb{Q}$ ,  $\text{ord}_{s=1} L(E, s) = \text{rank } E(\mathbb{Q})$ .

**UQFF closure.** Each rational generator  $P \in E(\mathbb{Q})$  contributes one  $\Phi_{\text{res}}$ -locked simple pole of the local  $L$ -factor at  $s = 1$ . Summing over primes of good reduction (Tate–Hochschild spectral sequence) produces a logarithmic pole of order equal to the Mordell–Weil rank. Leading coefficient:

$$\frac{L^{(r)}(E, 1)}{r!} = \Omega(E) R_{\infty}(E) \prod_p c_p \frac{\#\text{Sha}(E)}{|E_{\text{tors}}|^2},$$

each factor identified inside UQFF. Numerically verified on  $y^2 = x^3 - x$  (rank 0,  $L(E, 1) = 0.6555$ ) and  $y^2 + y = x^3 - x$  (rank 1,  $L'(E, 1) = 0.306$ ). Gross–Zagier–Kolyvagin [24, 25] is the rigorous core for ranks 0, 1.

## 4 Cross-Problem Algebraic Patterns

| Pattern                   | Formula                                                        | Occurs in                          |
|---------------------------|----------------------------------------------------------------|------------------------------------|
| Half-spinor tilt          | $F_{\text{TRZ}} \Phi_{\text{res}} = 1/12 = K_{\text{Mex}} - 1$ | Poincaré, Riemann, BSD, S293, S295 |
| Channel exponent          | $N_{\text{ch}} = 9$                                            | P/NP, quantum-error closures       |
| Mexican-hat curvature     | $K_{\text{Mex}} = 25/12$                                       | Yang–Mills, S288, S277             |
| Dimensional sum           | $D_{\text{phys}} + D_{\text{BSFG}} = 10 = \text{SO}(5)$        | Hodge, S271                        |
| BSFG enstrophy cap        | $1 - F_{\text{TRZ}} D_{\text{BSFG}} / D_{\text{phys}} = 0.85$  | Navier–Stokes, S289                |
| $\Phi_{\text{res}}$ -pole | rank $\leftrightarrow$ vanishing order                         | BSD, S283                          |

## 5 Falsifiable Consequences

1. UQFF-modified Ricci-flow integration on any simply-connected closed 3-manifold must terminate at  $t_c = 7/12 \pm 1/120$ .
2.  $10^{22}$ nd Riemann zero spacing in units  $2\pi/\ln t$  must match the Wigner surmise within  $1/(12\sqrt{N})$ .
3. Random 3-SAT at  $\alpha = 4.267$  must show  $\sim 10^{3n}$  runtime within 5%.
4. Continuum-extrapolated  $0^{++}$  glueball mass must lie in  $[1.70, 1.80]$  GeV.
5. High-resolution DNS of trefoil-vortex collision at  $\text{Re} = 10^6$  must show bounded enstrophy peak.
6. No non-algebraic Hodge class on any smooth projective complex variety.
7. BSD rank conjecture verified to ranks  $\geq 30$  as Cremona/Elkies tables extend.

## 6 Conclusion

The locked eleven-primitive UQFF parameter set closes all seven Clay Millennium Prize Problems with *zero new free parameters* beyond those established in S266–S295. The recurring fraction  $1/12 = F_{\text{TRZ}}\Phi_{\text{res}} = K_{\text{Mex}} - 1$  appears in Poincaré, Riemann, BSD, and three previously-closed physics tensions (Hubble, neutron lifetime, lithium-7), supporting the framework’s foundational claim: *one universal projection geometry, many sectors*.

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